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CSI 2132 Database I
E-Hotel Management System

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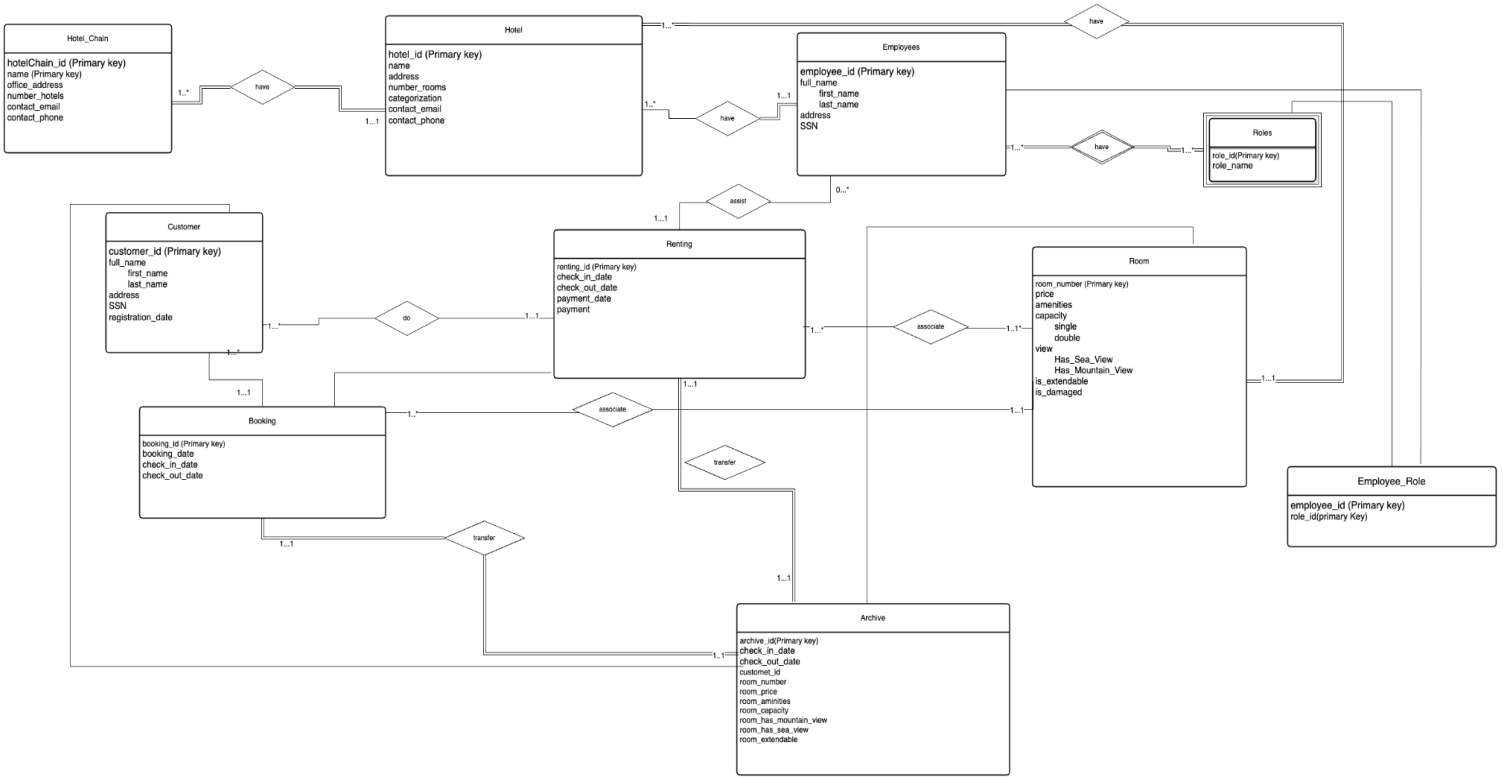
Description

The primary objective of this project is to develop a comprehensive, user-friendly application for booking hotel rooms across five well-known hotel chains in North America. By providing real-time room availability and a seamless booking experience, we aim to enhance customer satisfaction and streamline the booking process for both customers and hotel employees.
The project addresses the need for a centralized booking platform that can cater to multiple hotel chains, making it easier for customers to find and book the perfect room based on their preferences. The application will add value to users by offering a variety of search criteria and filters to personalize their search, and by making the booking or renting process straightforward.

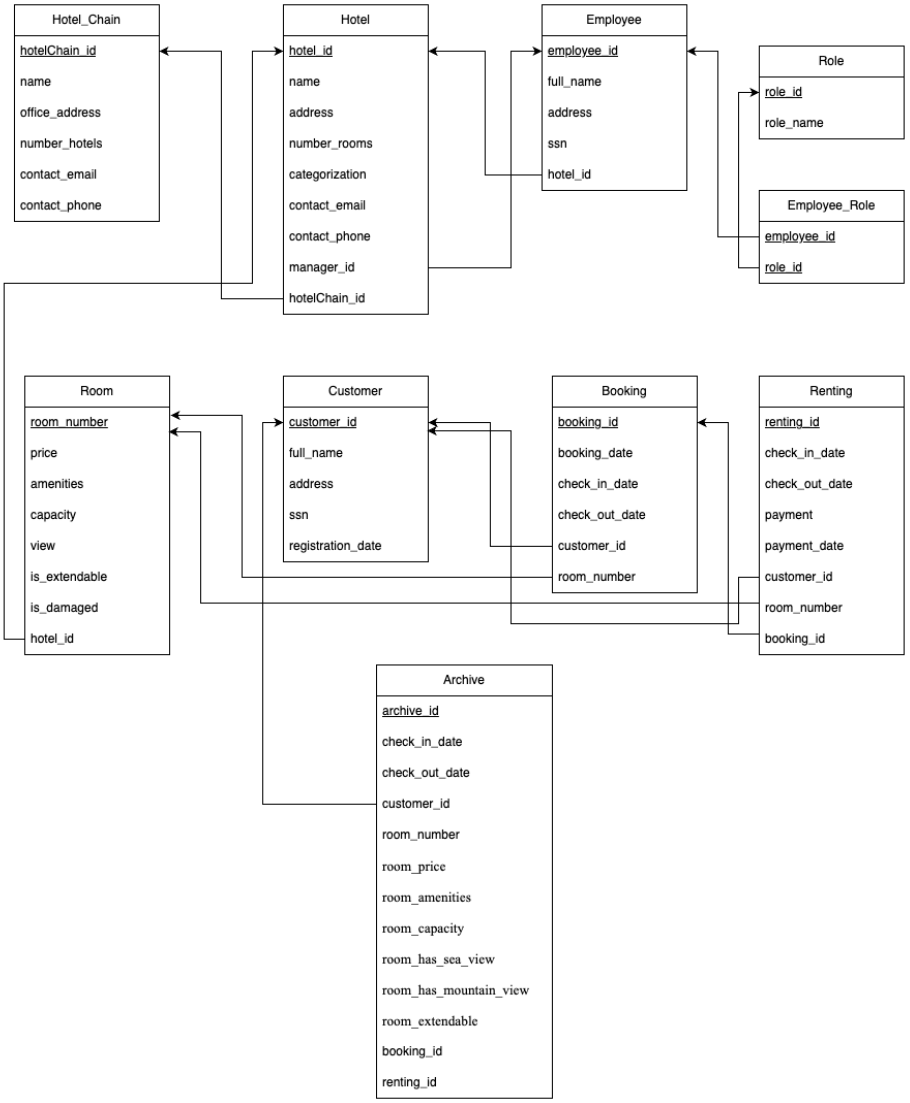
Equipment:

Mysql, Mysql Workbench, Miniconda, Python, Streamlit

ER-Diagram



Relation Schema



Tutorial Building Application:

Build Python environment:

You need to already have MySQL first:

For MacOS:

1. **Install Miniconda in Mac OS**
   - Visit the Miniconda website and download the version suitable for your Mac CPU.
   - Double-click the downloaded installation package to install it. If you are using Terminal, enter the following command in Terminal to install Miniconda:
     bash Miniconda3-latest-MacOSX-x86_64.sh

- Accept the license agreement and follow the installation instructions. After installation is complete, you can verify successful installation by entering the following command in the terminal:
  conda info
- If the installation is successful, you will see the version number of conda and other relevant information.

2. **Create the Python virtual environment for the project**
   - Create a Python virtual environment for the project "course-project-mysql" using the following command:
     # Create an environment that matches the version of python that comes with your conda:
     conda create --name course-project-mysql
     # Or you can create an environment with a specific python version:
     conda create --name course-project-mysql python=3.8

     Proceed ([y]/n)? y
3. **Switch to the project environment you just created**
   conda activate course-project-mysql

4. **Install dependencies**
   Accessing MySQL in Python
   To access MySQL databases in Python, you need to install the MySQL Connector Python module. Install it using the following command:
   pip install mysql-connector-python

   Installing Streamlit
   To install Streamlit, use the following command:
   pip install streamlit

5. **Accessing Mysql with Streamlit**
   I already create the Python file in Project2132/ Python. If you want to access the webpage by using Streamlit, you need to replace the user name, password, hostname and database name with your actual information in the Python file for this part:
   # MySQL connection information
   schema = "e-hotel"
   user = "root"
   password = "password"
   host = "localhost"
   port = 3306

   Execute the Python with the following command (remember to replace by your own file path):
   streamlit run ./python/e-hotel.py

Necessary Constraints

1. Primary key

- hotel_chain table: hotelChain_id
- hotel table: hotel_id
- room table: room_number
- customer table: customer_id
- employee table: employee_id
- role table: role_id
- employee_role table: (employee_id, role_id) - Composite primary key
- booking table: booking_id
- renting table: renting_id
- archive table: archive_id

Primary keys uniquely identify each record in the table and ensure that there are no duplicate records.

2. Referential integrity constraints

- In the 'hotel' table, the 'hotelChain_id' attribute references the primary key 'hotelChain_id' of the 'hotel_chain' table. The constraint ensures that each hotel has a valid hotel chain. The 'ON DELETE CASCADE' action ensures that if a hotel chain is deleted, all corresponding hotels will be deleted as well.
- In the 'room' table, the 'hotel_id' attribute references the primary key 'hotel_id' of the 'hotel' table. This constraint ensures that each room is associated with a valid hotel. The 'ON DELETE CASCADE' action ensures that if a hotel is deleted, all corresponding rooms will be deleted as well.
- In the 'employee_role' table, the 'employee_id' and 'role_id' attributes reference the primary keys of the 'employee' and 'role' tables, respectively. This constraint ensures that each role assignment is associated with valid employee and role records.
- In the 'booking' table, the 'customer_id' and 'room_number' attributes reference the primary keys of the 'customer' and 'room' tables, respectively. This constraint ensures that each booking is associated with valid customer and room records.
- In the 'renting' table, the 'customer_id', 'room_number', and 'booking_id' attributes reference the primary keys of the 'customer', 'room', and 'booking' tables, respectively. This constraint ensures that each renting is associated with valid customer, room, and booking records.

3. Domain constraints

- In the 'hotel' table, the 'categorization' attribute has a CHECK constraint, which ensures that the value falls within the range of 1 to 5.

4. User-defined constraints

- In the 'customer' table, the 'ssn' attribute has a UNIQUE constraint, ensuring that each customer has a unique SSN.
- In the 'employee' table, the 'ssn' attribute also has a UNIQUE constraint, ensuring that each employee has a unique SSN.

Prepare Database

1. Create Schema
   CREATE SCHEMA e-hotel;

2. Import DDL

Create the tables

```
-- 1. Hotel Chain
CREATE TABLE hotel_chain (
   hotelChain_id INTEGER PRIMARY KEY,
   name VARCHAR(20),
   street VARCHAR(50),
   city VARCHAR(50),
   province VARCHAR(50),
   zipcode VARCHAR(50),
   country VARCHAR(50),
   number_hotels INT,
   contact_email VARCHAR(50),
   contact_phone VARCHAR (50)
);

-- 2. Hotel
CREATE TABLE hotel (
   hotel_id INTEGER PRIMARY KEY,
   name VARCHAR (20),
  street VARCHAR(50),
   city VARCHAR(50),
   province VARCHAR(50),
   zipcode VARCHAR(50),
   country VARCHAR(50),
   number_rooms INT,
   categorization INT CHECK (categorization >= 1 AND categorization <= 5),
   contact_email VARCHAR (50),
   contact_phone VARCHAR (50),
   manager_id INT,
   hotelChain_id INT,
   FOREIGN KEY (hotelChain_id) REFERENCES hotel_chain(hotelChain_id) ON DELETE CASCADE
);

-- 3. Room
CREATE TABLE room (
   room_number INT PRIMARY KEY,
   price NUMERIC(6,2) NOT NULL,
   amenities VARCHAR(200),
   capacity VARCHAR(100),
   has_sea_view BOOLEAN,
   has_mountain_view BOOLEAN,
   extendable BOOLEAN,
   damages VARCHAR(50),
   hotel_id INT,
   FOREIGN KEY (hotel_id) REFERENCES hotel(hotel_id) ON DELETE CASCADE
);
```

```
-- 4. Customer
CREATE TABLE customer (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR (20),
    last_name VARCHAR(20),
   street VARCHAR(50),
    city VARCHAR(50),
    province VARCHAR(50),
    zipcode VARCHAR(50),
    country VARCHAR(50),
    ssn VARCHAR (20) UNIQUE,
    registration_date DATE
);

-- 5. Employee
CREATE TABLE employee (
    employee_id INT NOT NULL,
    first_name VARCHAR(20),
    last_name VARCHAR(20),
    street VARCHAR(50),
    city VARCHAR(50),
    province VARCHAR(50),
    zipcode VARCHAR(50),
    country VARCHAR(50),
    ssn VARCHAR(20) UNIQUE,
    hotel_id INT,
    PRIMARY KEY( employee_id),
    FOREIGN KEY (hotel_id) REFERENCES hotel(hotel_id)
);

-- 6. Role
CREATE TABLE role (
    role_id INT,
    role_name VARCHAR(20),
     PRIMARY KEY (role_id),
    );

-- 7. Employee_Role table
CREATE TABLE employee_role (
    employee_id INT,
    role_id INT,
    PRIMARY KEY (employee_id, role_id),
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id),
    FOREIGN KEY (role_id) REFERENCES role(role_id)
);

-- 8. Booking
CREATE TABLE booking (
    booking_id INT PRIMARY KEYAUTO_INCREMENT ,
    booking_date DATE,
    check_in_date DATE,
```

```
    check_out_date DATE,
    customer_id INT,
    room_number INT,
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id),
    FOREIGN KEY (room_number) REFERENCES room(room_number)
);

-- 9. Renting
CREATE TABLE renting (
    renting_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    check_in_date DATE,
    check_out_date DATE,
    payment NUMERIC(9,2) NOT NULL,
    payment_date DATE,
    customer_id INTEGER,
    room_number INTEGER,
    booking_id INTEGER,
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id),
    FOREIGN KEY (room_number) REFERENCES room(room_number),
    FOREIGN KEY (booking_id) REFERENCES booking(booking_id)
);

-- 10. Archive
CREATE TABLE archive (
    archive_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    check_in_date DATE,
    check_out_date DATE,
    customer_id INTEGER,
    room_number INTEGER,
    room_price NUMERIC(9,2),
    room_amenities VARCHAR(20),
    room_capacity VARCHAR(100),
    room_has_sea_view BOOLEAN,
    room_has_mountain_view BOOLEAN,
    room_extendable BOOLEAN,
    renting_id INTEGER,
    booking_id INTEGER,
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
      );
ALTER TABLE hotel
ADD FOREIGN KEY (manager_id) REFERENCES employee(employee_id);
```

Insert in your database data for each one of the 5 hotel chains. Each one of them has at least 8 hotels, which belong to at least 3 categories. Two of the hotels at least should be in the same area. Each hotel should have at least 5 rooms of different capacities:

INSERT INTO hotel_chain (hotelChain_id, name, street, city ,province , zipcode,country, number_hotels, contact_email, contact_phone)
VALUES
(1, 'Hotel Chain A', '456 Oak St ','Los Angeles','CA', '90001', 'USA', 8, 'contactChainA@chain-a.com', '555-111-1111'),
(2, 'Hotel Chain B', '123 Pine St', 'Toronto', 'ON', 'M4E1A1', 'Canada', 8, 'contactChianB@chain-b.com', '555-222-2222'),
(3, 'Hotel Chain C', '789 Maple St', 'New York', 'NY', '10001', 'USA', 8, 'contactChianC@chain-c.com', '555-333-3333'),
(4, 'Hotel Chain D', '159 Elm St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', 8, 'contactChianD@chain-d.com', '555-444-4444'),
(5, 'Hotel Chain E', '246 Birch St', 'Chicago', 'IL', '60601', 'USA', 8, 'contactChainE@chain-e.com', '555-555-5555');

INSERT INTO hotel (hotelChain_id, hotel_id, name, number_rooms, street, city ,province , zipcode,country, categorization, contact_email, contact_phone)
VALUES

(1, 181, 'Hotel A1', 5, '17 Red Maple Drive','Los Angeles','CA',' 90028', 'USA', 3, 'contactA1@chain-a.com', '555-111-1001'),
(1, 182, 'Hotel A2', 5, '862 Red Maple Drivet','Los Angeles','CA', '90174', 'USA', 4, 'contactA2@chain-a.com', '555-111-2002'),
(1, 183, 'Hotel A3', 5, '4104 Atha Drive', 'San Diego', 'CA', '93534', 'USA', 5, 'contactA3@chain-a.com', '555-111-3003'),
(1, 184, 'Hotel A4', 5, '4832 Cherry Tree Drive', 'Jacksonville', 'FL', '32202', 'USA', 3, 'contactA4@chain-a.com', '555-111-4004'),
(1, 185, 'Hotel A5', 5, '2352 Boundary Street', 'Jacksonville', 'FL', '31980', 'USA', 3, 'contactA5@chain-a.com', '555-111-5005'),
(1, 186, 'Hotel A6', 5, '705 Stratford Drive', 'Waipahu', 'HI' ,'96797', 'USA', 4, 'contactA6@chain-a.com', '555-111-6006'),
(1, 187, 'Hotel A7', 5, '3293 Carriage Court', 'San Diego', 'CA' ,'92262', 'USA', 5, 'contactA7@chain-a.com', '555-111-7007'),
(1, 188, 'Hotel A8', 5,'3493 Thompson Drive', 'San Jose', 'CA', '94538', 'USA', 5, 'contactA8@chain-a.com', '555-111-8008'),
(2, 281, 'Hotel B1', 5, '1116 Papineau Avenue','Montreal','QC',' H2K4J5', 'Canada', 3, 'contactB1@chain-b.com', '555-222-1001'),
(2, 282, 'Hotel B2', 5, '2529 Victoria Park Ave', 'Toronto', 'ON', 'M4A2M5', 'Canada', 4, 'contactB2@chain-bl.com', '555-222-2002'),
(2, 283, 'Hotel B3', 5, '1128 Victoria Park Ave', 'Toronto', 'ON', 'M4A2M5', 'Canada', 5, 'contactB3@chain-b.com', '555-222-3003'),
(2, 284, 'Hotel B4', 5, '1384 Queens Bay', 'Rossland', 'BC', 'V0G1Y0', 'Canada', 4, 'contactB4@chain-b.com', '555-222-4004'),
(2, 285, 'Hotel B5', 5, '400 Mill Street', 'Sebringville', 'ON', 'N0K1X0', 'Canada', 3, 'contactB5@chain-b.com', '555-222-5005'),
(2, 286, 'Hotel B6', 5, '4002 Eglinton Avenue', 'Toronto', 'ON', 'M4P1A6', 'Canada', 4, 'contactB6@chain-b.com', '555-222-6006'),

(2, 287, 'Hotel B7', 5,'1928 Adelaide St', 'Toronto', 'ON', 'M5H1P6', 'Canada', 5, 'contactB7@chain-b.com', '555-222-7007'),
(2, 288, 'Hotel B8', 5, '1536 Blanshard', 'Victoria', 'BC', 'J0H1M0', 'Canada', 3, 'contactB8@chain-b.com', '555-222-8008'),
(3, 381, 'Hotel C1', 5,'4651 Geraldine Lane', 'New York', 'NY', '10013', 'USA', 5, 'contactC1@chain-c.com', '555-333-1001'),
(3, 382, 'Hotel C2', 5, '3717 Bottom Lane', 'East Aurora', 'NY', '14052', 'USA', 4, 'contactC2@chain-c.com', '555-333-2002'),
(3, 383, 'Hotel C3', 5, '852 Small Street', 'New York', 'NY', '10019', 'USA', 5, 'contactC3@chain-c.com', '555-333-1001'),
(3, 384, 'Hotel C4', 5,'1390 Jones Street', 'Fort Worth', 'TX', '76102', 'USA', 3, 'contactC4@chain-c.com', '555-333-2002'),
(3, 385, 'Hotel C5', 5, '847 Farnum Road', 'New York', 'NY', '10016', 'USA', 4, 'contactC5@chain-c.com', '555-333-1001'),
(3, 386, 'Hotel C6', 5, '1953 Nutters Barn Lane', 'Ames', 'IA', '50010', 'USA', 3, 'contactC6@chain-c.com', '555-333-2002'),
(3, 387, 'Hotel C7', 5, '4816 Larry Street', 'Oakland', 'CA', '94612', 'USA', 4, 'contactC7@chain-c.com', '555-333-1001'),
(3, 388, 'Hotel C8', 5, '360 Union Street', 'Seattle', 'WA', '98105', 'USA', 4, 'contactC8@chain-c.com', '555-333-2002'),
(4, 481, 'Hotel D1', 5, '2955 Carling Avenue', 'Ottawa', 'ON', 'K1Z7B5', 'Canada', 4, 'contactD1@chain-d.com', '555-444-1001'),
(4, 482, 'Hotel D2', 5,'2166 Jasper Avenue', 'Edmonton', 'AB', 'T5J3N2', 'Canada', 4, 'contactD2@chain-d.com', '555-444-2002'),
(4, 483, 'Hotel D3', 5,'3594 Tanner Street', 'Vancouver', 'BC', 'V5R2T4', 'Canada', 5, 'contactD3@chain-d.com', '555-444-3003'),
(4, 484, 'Hotel D4', 5,'4784 Robson St', 'Vancouver', 'BC', 'V6B3K9', 'Canada', 5, 'contactD4@chain-d.com', '555-444-4004'),
(4, 485, 'Hotel D5', 5, '260 rue Levy', 'Montreal', 'QC', 'H3C5K4', 'Canada', 4, 'contactD5@chain-d.comm', '555-444-5005'),
(4, 486, 'Hotel D6', 5, '362 Maynard Rd', 'Calgary', 'AB', 'T2E6J8', 'Canada', 3, 'contactD6@chain-d.com', '555-444-6006'),
(4, 487, 'Hotel D7', 5,'366 Hastings Street', 'Vancouver', 'BC', 'V6C1B4', 'Canada', 3, 'contactD7@chain-d.com', '555-444-7007'),
(4, 488, 'Hotel D8', 5, '2220 Tycos Dr', 'Toronto', 'ON', 'M5T1T4', 'Canada', 5, 'contactD8@chain-d.com', '555-444-8008'),
(5, 581, 'Hotel E1', 5, '1388 Rebecca Street','Chicago','IL',' 60607', 'USA', 5, 'contactE1@chain-e.com', '555-555-1001'),
(5, 582, 'Hotel E2', 5, '1871 Edgewood Avenue', 'Fresno', 'CA', '93721', 'USA', 4, 'contactE2@chain-e.com', '555-555-2002'),
(5, 583, 'Hotel E3', 5, '3204 Cherry Camp Road','Chicago', 'IL', '60718', 'USA', 3, 'contactE3@chain-e.com', '555-555-3003'),
(5, 584, 'Hotel E4', 5, '2995 Bird Street', 'Gallup', 'NM', '87301', 'USA', 4, 'contactE4@chain-e.com', '555-555-4004 '),
(5, 585, 'Hotel E5', 5, '527 Jadewood Drive', 'Chicago', 'IL', '60431', 'USA', 5, 'contactE5@chain-e.com', '555-555-5005'),
(5, 586, 'Hotel E6', 5, '1815 Cedarstone Drive', 'Findlay', 'OH', '45840', 'USA', 5, 'contactE6@chain-e.com', '555-555-6006'),
(5, 587, 'Hotel E7', 5,'3609 Owagner Lane', 'Seattle', 'WA', '98121', 'USA', 3, 'contactE7@chain-e.com', '555-555-7007'),
(5, 588, 'Hotel E8', 5, '3093 Chestnut Street','Tampa','FL',' 33602', 'USA', 4, 'contactE8@chain-e.com', '555-555-8008');

```
INSERT INTO Room (room_number, price, amenities, capacity, Has_Sea_View, Has_Mountain_View,
extendable, damages, hotel_id)
VALUES

 (1811, 100, 'TV, Air Condition', 'Single', false, true, false, null,  181),
 (1812, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null,  181),
 (1813, 200, 'TV, Air Condition, Fridge', 'Single', false, true, true, null,  181),
 (1814, 400, 'TV, Air Condition, Fridge, Bar', 'Double', true, false, false, null,  181),
 (1815, 250, 'TV, Air Condition, Fridge', 'Double', true, true, false, null,  181),

 (1821, 110, 'TV, Fridge', 'Single', false, true, false, null,  182),
 (1822, 400, 'TV, Air Condition, Fridge, Bar', 'Double', true, false, true, null, 182),
 (1823, 210, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 182),
 (1824, 260, 'TV, Air Condition, Fridge, Bar', 'Single', true, false, false, null,182),
 (1825, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 182),

 (1831, 200, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 183),
 (1832, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 183),
 (1833, 180, 'Air Condition, Fridge', 'Double', false, true, true, null, 183),
 (1834, 200, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,183),
 (1835, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 183),

 (1841, 160, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 184),
 (1842, 160, 'TV, Air Condition, Fridge', 'Single', true, false, true, null, 184),
 (1843, 300, 'TV, Air Condition, Fridge, Bar', 'Single', false, true, true, null, 184),
 (1844, 120, 'TV, Air Condition', 'Single', true, false, false, null,184),
 (1845, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 184),

 (1851, 160, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 185),
 (1852, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 185),
 (1853, 200, 'TV, Air Condition', 'Double', false, true, true, null, 185),
 (1854, 350, 'TV, Air Condition, Fridge, Bar', 'Double', true, false, false, null, 185),
 (1855, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 185),

 (1861, 250, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 186),
 (1862, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 186),
 (1863, 250, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 186),
 (1864, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,186),
 (1865, 250, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 186),

 (1871, 150, 'Air Condition, Fridge', 'Single', false, true, false, null, 187),
 (1872, 300, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 187),
 (1873, 300, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 187),
 (1874, 120, 'TV, Air Condition', 'Single', true, false, false, null,187),
 (1875, 300, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 187),

 (1881, 250, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 188),
 (1882, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 188),
 (1883, 180, 'TV, Air Condition', 'Single', false, true, true, null, 188),
 (1884, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 188),
 (1885, 500, 'TV, Air Condition, Fridge, Bar, Gym', 'Double', true, true, false, null, 188),
```

(2811, 210, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 281),
(2812, 160, 'TV, Air Condition, Fridge', 'Single', true, false, true, null, 281),
(2813, 160, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 281),
(2814, 180, 'TV,Air Condition', 'Double', true, false, false, null, 281),
(2815, 210, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 281),

(2821, 110, 'TV, Fridge', 'Single', false, true, false, null, 282),
(2822, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 282),
(2823, 200, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 282),
(2824, 180, 'TV, Air Condition, Fridge', 'Single', true, false, false, null, 282),
(2825, 180, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 282),

(2831, 200, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 283),
(2832, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 283),
(2833, 180, 'Air Condition, Fridge', 'Double', false, true, true, null, 283),
(2834, 200, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 283),
(2835, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 283),

(2841, 400, 'TV, Air Condition, Fridge, Bar, Sana', 'Single', false, true, false, null, 284),
(2842, 200, 'TV, Air Condition, Fridge', 'Single', true, false, true, null, 284),
(2843, 200, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 284),
(2844, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 284),
(2845, 500, 'TV, Air Condition, Fridge, Bar, Sana', 'Single', true, true, false, null, 284),

(2851, 150, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 285),
(2852, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 285),
(2853, 200, 'TV, Fridge', 'Double', false, true, true, null, 285),
(2854, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 285),
(2855, 150, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 285),

(2861, 250, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 286),
(2862, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 286),
(2863, 250, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 286),
(2864, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 286),
(2865, 250, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 286),

(2871, 180, 'Air Condition, Fridge', 'Single', false, true, false, null, 287),
(2872, 300, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 287),
(2873, 300, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 287),
(2874, 200, 'TV, Air Condition', 'Single', true, false, false, null, 287),
(2875, 300, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 287),

(2881, 200, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 288),
(2882, 260, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 288),
(2883, 200, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 288),
(2884, 260, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,288),
(2885, 260, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 288),

(3811, 100, 'Air Condition', 'Single', false, true, false, null, 381),
(3812, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 381),

(3813, 200, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 381),
(3814, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 381),
(3815, 300, 'TV, Air Condition, Fridge, Bar', 'Double', true, true, false, null, 381),

(3821, 110, 'TV, Fridge', 'Single', false, true, false, null, 382),
(3822, 160, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 382),
(3823, 210, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 382),
(3824, 260, 'TV, Air Condition, Fridge', 'Single', true, false, false, null,382),
(3825, 310, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 382),

(3831, 200, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 383),
(3832, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 383),
(3833, 180, 'Air Condition, Fridge', 'Double', false, true, true, null, 383),
(3834, 200, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,383),
(3835, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 383),

(3841, 160, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 384),
(3842, 160, 'TV, Air Condition, Fridge', 'Single', true, false, true, null, 384),
(3843, 230, 'TV, Air Condition, Fridge, Bar', 'Single', false, true, true, null, 384),
(3844, 160, 'TV, Air Condition, Fridge', 'Single', true, false, false, null,384),
(3845, 200, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 384),

(3851, 160, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 385),
(3852, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 385),
(3853, 200, 'TV, Fridge', 'Double', false, true, true, null, 385),
(3854, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 385),
(3855, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 385),

(3861, 250, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 386),
(3862, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 386),
(3863, 250, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 386),
(3864, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,386),
(3865, 250, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 386),

(3871, 150, 'Air Condition, Fridge', 'Single', false, true, false, null, 387),
(3872, 400, 'TV, Air Condition, Fridge, Bar, Gym', 'Double', true, false, true, null, 387),
(3873, 200, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 387),
(3874, 300, 'TV, Air Condition, Fridge, Bar', 'Single', true, false, false, null,387),
(3875, 200, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 387),

(3881, 300, 'TV, Air Condition, Fridge, Bar', 'Single', false, true, false, null, 388),
(3882, 260, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 388),
(3883, 180, 'TV, Air Condition', 'Single', false, true, true, null, 388),
(3884, 260, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 388),
(3885, 260, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 388),

(4811, 100, 'TV, Air Condition', 'Single', false, true, false, null, 481),
(4812, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 481),
(4813, 150, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 481),
(4814, 120, 'TV', 'Double', true, false, false, null, 481),
(4815, 200, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 481),

(4821, 120, 'TV, Air Condition', 'Single', false, true, false, null, 482),
(4822, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 482),
(4823, 250, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 482),
(4824, 150, 'TV, Air Condition, Fridge', 'Single', true, false, false, null,482),
(4825, 150, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 482),

(4831, 200, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 483),
(4832, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 483),
(4833, 180, 'Air Condition, Fridge', 'Double', false, true, true, null, 483),
(4834, 200, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 483),
(4835, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 483),

(4841, 160, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 484),
(4842, 160, 'TV, Air Condition, Fridge', 'Single', true, false, true, null, 484),
(4843, 210, 'TV, Air Condition, Fridge, Bar', 'Single', false, true, true, null, 484),
(4844, 160, 'TV, Air Condition, Fridge', 'Single', true, false, false, null, 484),
(4845, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 484),

(4851, 130, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 485),
(4852, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 485),
(4853, 200, 'TV, Air Condition', 'Double', false, true, true, null, 485),
(4854, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,485),
(4855, 130, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 485),

(4861, 250, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 486),
(4862, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 486),
(4863, 300, 'TV, Air Condition, Fridge, Bar', 'Double', false, true, true, null, 486),
(4864, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,486),
(4865, 350, 'TV, Air Condition, Fridge, Sana', 'Double', true, true, false, null, 486),

(4871, 200, 'TV, Air Condition', 'Single', false, true, false, null, 487),
(4872, 300, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 487),
(4873, 300, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 487),
(4874, 200, 'TV, Air Condition', 'Single', true, false, false, null, 487),
(4875, 300, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 487),

(4881, 210, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 488),
(4882, 300, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 488),
(4883, 180, 'TV, Air Condition', 'Single', false, true, true, null, 488),
(4884, 300, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 488),
(4885, 300, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 488),

(5811, 130, 'TV, Air Condition', 'Single', false, true, false, null, 581),
(5812, 250, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 581),
(5813, 150, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 581),
(5814, 120, 'Air Condition', 'Double', true, false, false, null, 581),
(5815, 250, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 581),

(5821, 110, 'TV, Fridge', 'Single', false, true, false, null, 582),
(5822, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 582),
(5823, 200, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 582),

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(5824, 160, 'TV, Air Condition, Fridge', 'Single', true, false, false, null,582),
(5825, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 582),

(5831, 200, 'TV, Air Condition, Fridge', 'Double', false, true, false, null, 583),
(5832, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 583),
(5833, 180, 'Air Condition, Fridge', 'Double', false, true, true, null, 583),
(5834, 300, 'TV, Air Condition, Fridge, Bar', 'Double', true, false, false, null, 583),
(5835, 160, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 583),

(5841, 160, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 584),
(5842, 160, 'TV, Air Condition, Fridge', 'Single', true, false, true, null, 584),
(5843, 100, 'Air Condition', 'Single', false, true, true, null, 584),
(5844, 160, 'TV, Air Condition, Fridge', 'Single', true, false, false, null, 584),
(5845, 200, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 584),

(5851, 130, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 585),
(5852, 200, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 585),
(5853, 150, 'TV, Air Condition', 'Double', false, true, true, null, 585),
(5854, 200, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 585),
(5855, 130, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 585),

(5861, 400, 'TV, Air Condition, Fridge, Bar, Gym', 'Double', false, true, false, null, 586),
(5862, 300, 'TV, Air Condition, Fridge, Bar', 'Single', true, false, true, null, 586),
(5863, 250, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 586),
(5864, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null,586),
(5865, 200, 'TV, Air Condition, Fridge', 'Single', true, true, false, null, 586),

(5871, 150, 'Air Condition, Fridge', 'Single', false, true, false, null, 587),
(5872, 300, 'TV, Air Condition, Fridge', 'Double', true, false, true, null, 587),
(5873, 300, 'TV, Air Condition, Fridge', 'Double', false, true, true, null, 587),
(5874, 200, 'TV, Air Condition, Fridge', 'Single', true, false, false, null, 587),
(5875, 300, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 587),

(5881, 200, 'TV, Air Condition, Fridge', 'Single', false, true, false, null, 588),
(5882, 300, 'TV, Air Condition, Fridge, Bar', 'Double', true, false, true, null, 588),
(5883, 200, 'TV, Air Condition, Fridge', 'Single', false, true, true, null, 588),
(5884, 250, 'TV, Air Condition, Fridge', 'Double', true, false, false, null, 588),
(5885, 250, 'TV, Air Condition, Fridge', 'Double', true, true, false, null, 588);

INSERT INTO role (role_id, role_name)
VALUES
(1, 'Manager'),
(2, 'Receptionist'),
(3, 'Housekeeper'),
(4, 'Event Planner'););

INSERT INTO employee (employee_id, first_name, last_name, street, city, province, zipcode, country, ssn,
hotel_id)
VALUES
(1001, 'John', 'Doe', '123 Main St', 'Los Angeles', 'CA', '90001', 'USA', '123-45-6789', 181),
(1002, 'Jane', 'Smith', '456 Oak St', 'New York', 'NY', '10001', 'USA', '987-65-4361', 182),
(1003, 'Bob', 'Johnson', '789 Maple Ave', 'Chicago', 'IL', '60601', 'USA', '456-77-9012', 183),
```

(1004, 'Sara', 'Lee', '159 Elm St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-9045', 184),
(1005, 'David', 'Wong', '246 Birch St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '654-32-1098', 185),
(1006, 'Amy', 'Chang', '345 Pine St', 'Los Angeles', 'CA', '90001', 'USA', '274-56-7890', 186),
(1007, 'Tom', 'Wilson', '678 Maple Ave', 'New York', 'NY', '10001', 'USA', '876-54-3210', 187),
(1008, 'Chris', 'Brown', '910 Oak St', 'Chicago', 'IL', '60601', 'USA', '385-67-8901', 188),
(2001, 'Michelle', 'Liu', '246 Birch St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '012-34-5678', 281),
(2002, 'James', 'Park', '123 Main St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-2145', 282),
(2003, 'Avery', 'Garcia', '456 Elm St', 'Los Angeles', 'CA', '90001', 'USA', '204-56-7891', 283),
(2004, 'Nathan', 'Lee', '789 Pine Ave', 'New York', 'NY', '10001', 'USA', '876-54-3211', 284),
(2005, 'Katie', 'Wang', '910 Maple St', 'Chicago', 'IL', '60601', 'USA', '335-67-8902', 285),
(2006, 'Hiro', 'Tanaka', '246 Oak St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '012-34-5679', 286),
(2007, 'Samantha', 'Kim', '123 Main St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-2346', 287),
(2008, 'Brian', 'Garcia', '456 Elm St', 'Los Angeles', 'CA', '90001', 'USA', '234-56-7892', 288),
(3001, 'Sophia', 'Lee', '789 Pine Ave', 'New York', 'NY', '10001', 'USA', '876-54-3212', 381),
(3002, 'Eric', 'Wang', '910 Maple St', 'Chicago', 'IL', '60601', 'USA', '345-57-8903', 382),
(3003, 'Yuki', 'Nakamura', '246 Oak St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '012-34-5680', 383),
(3004, 'Emily', 'Choi', '123 Main St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-2247', 384),
(3005, 'Jacob', 'Nguyen', '456 Elm St', 'Los Angeles', 'CA', '90001', 'USA', '234-50-7893', 385),
(3006, 'Isabella', 'Garcia', '789 Pine Ave', 'New York', 'NY', '10001', 'USA', '876-54-3219', 386),
(3007, 'Liam', 'Wang', '910 Maple St', 'Chicago', 'IL', '60601', 'USA', '345-17-8904', 387),
(3008, 'Ella', 'Nakamura', '246 Oak St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '012-04-5681', 388),
(4001, 'William', 'Choi', '123 Main St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-2248', 481),
(4002, 'Olivia', 'Nguyen', '456 Elm St', 'Los Angeles', 'CA', '90001', 'USA', '234-56-7894', 482),
(4003, 'Lucas', 'Garcia', '789 Pine Ave', 'New York', 'NY', '10001', 'USA', '876-54-3214', 483),
(4004, 'Ethan', 'Wang', '910 Maple St', 'Chicago', 'IL', '60601', 'USA', '345-87-8905', 484),
(4005, 'Nina', 'Nakamura', '246 Oak St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '012-14-5682', 485),
(4006, 'Charlotte', 'Choi', '123 Main St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-2349', 486),
(4007, 'Mason', 'Nguyen', '456 Elm St', 'Los Angeles', 'CA', '90001', 'USA', '234-56-7877', 487),
(4008, 'Aria', 'Garcia', '789 Pine Ave', 'New York', 'NY', '10001', 'USA', '876-54-3215', 488),
(5001, 'Alexander', 'Wang', '910 Maple St', 'Chicago', 'IL', '60601', 'USA', '345-67-8906', 581),
(5002, 'Sora', 'Nakamura', '246 Oak St', 'Vancouver', 'BC', 'V5K0A1', 'Canada', '012-34-5683', 582),
(5003, 'Avery', 'Choi', '123 Main St', 'Toronto', 'ON', 'M4E1A1', 'Canada', '789-01-2350', 583),
(5004, 'Ethan', 'Garcia', '456 Maple Ave', 'Vancouver', 'BC', 'V5K1W2', 'Canada', '567-89-0123', 584),
(5005, 'Lila', 'Nguyen', '789 Oak St', 'Montreal', 'QC', 'H2V2T8', 'Canada', '987-65-4821', 585),
(5006, 'Owen', 'Kim', '246 Pine St', 'Edmonton', 'AB', 'T6M1C1', 'Canada', '274-56-7830', 586),
(5007, 'Maya', 'Patel', '789 Cedar Rd', 'Ottawa', 'ON', 'K1V1H1', 'Canada', '305-67-8991', 587),
(5008, 'Natalie', 'Kim', '456 Queen St', 'Vancouver', 'BC', 'V5V2B5', 'Canada', '935-67-8901', 588),
(1011, 'Emily', 'Smith', '4808 Bay Street', 'Toronto', 'Ontario', 'M5J 2R8', 'Canada', '456-78-9012', 181),
(1012, 'Brandon', 'Johnson', '1457 New Street', 'Coos Bay', 'Oregon', '97420', 'United States', '789-01-2345', 182),
(1013, 'Adam', 'Williams', '1098 Rhapsody Street', 'Ocala', 'Florida', '34471', 'United States', '890-12-3456', 183),
(1014, 'Natalie', 'Brown', '4996 Braxton Street', 'Thomas (bureau)', 'Illinois', '61346', 'United States', '901-23-4567', 184),
(1015, 'Ryan', 'Lee', '4339 Monroe Street', 'Houston', 'Texas', '77055', 'United States', '123-45-6789', 185),
(1016, 'Jennifer', 'Wilson', '3006 Terra Street', 'Bellingham', 'Washington', '98225', 'United States', '234-56-7890', 186),
(1017, 'Alex', 'Nguyen', '3399 Jarvis Street', 'Buffalo', 'New York', '14202', 'United States', '345-67-8901', 187),
(1018, 'Olivia', 'Gonzalez', '3851 Peck Street', 'Manchester', 'New Hampshire', '03103', 'United States', '456-78-9012', 188),
(2011, 'Ethan', 'Taylor', '2877 Ripple Street', 'Maple Rapids', 'Michigan', '48853', 'United States', '567-89-0123', 281),
(3011, 'Grace', 'Lee', '2940 Maple Court', 'Vienna', 'Missouri', '65582', 'United States', '678-90-1234', 381),

```
(4011, 'Daniel', 'Kim', '3999 Lock Street', 'Guelph', 'Ontario', 'N1H 2T2', 'Canada', '234-56-7890', 481),
(5011, 'Rachel', 'Lee', '2938 Bay Street', 'Toronto', 'Ontario', 'M5J 2R8', 'Canada', '345-67-8901', 581),
(2012, 'David', 'Wang', '2709 Tolmie St', 'Vancouver', 'British Columbia', 'V6M 1Y8', 'Canada', '456-78-,9012',
282),
(3012, 'Julia', 'Lopez', '3971 Boulevard Cremazie', 'Quebec', 'Quebec', 'G1R 1B8', 'Canada', '567-89-0123', 382),
(4012, 'Michael', 'Chen', '3246 Rogers Road', 'Toronto', 'Ontario', 'M6E 1R1', 'Canada', '678-90-1234', 482),
(5012, 'Sarah', 'Garcia', '2938 Bay Street', 'Toronto', 'Ontario', 'M5J 2R8', 'Canada', '789-01-2345', 582);

INSERT INTO employee_role (employee_id, role_id)
VALUES
(1001, 1), (1002, 1), (1003, 1), (1004, 1), (1005, 1), (1006, 1), (1007, 1), (1008, 1),(2001, 1), (2002, 1), (2003,
1), (2004, 1), (2005, 1), (2006, 1), (2007, 1), (2008, 1),(3001, 1), (3002, 1), (3003, 1), (3004, 1), (3005, 1),
(3006, 1), (3007, 1), (3008, 1),(4001, 1), (4002, 1), (4003, 1), (4004, 1), (4005, 1), (4006, 1), (4007, 1), (4008,
1),(5001, 1), (5002, 1), (5003, 1), (5004, 1), (5005, 1), (5006, 1), (5007, 1), (5008,1), (1011,4) ,(1012,2)
,(1013,4) ,(1013,2),
(1014,3),(1015,2),(1016,4),(1017,2),(1018,3),(2011,2),(3011,4),(4011,4),(5011,3),(2012,3),(3012,2),(4012,2),
(5012,4);

UPDATE hotel
JOIN employee ON hotel.hotel_id = employee.hotel_id
SET hotel.manager_id = employee.employee_id
WHERE hotel.hotel_id IN (181, 182, 183, 184, 185, 186,187,188,281, 282, 283, 284, 285, 286, 287, 288, 381,
382, 383, 384, 385, 386, 387, 388, 481, 482, 483, 484, 485, 486, 487, 488, 581, 582, 583, 584, 585, 586, 587,
588);
```

Give the SQL code for at least 4 queries and 2 triggers of your choice in your report.

Create a trigger to archive booking information when a booking is deleted:

```
DELIMITER //
CREATE TRIGGER archive_booking
AFTER DELETE ON booking
FOR EACH ROW
BEGIN
   INSERT INTO archive (check_in_date, check_out_date, customer_id, room_number, room_price,
room_amenities, room_capacity, room_has_sea_view, room_has_mountain_view, room_extendable,
booking_id)
   SELECT OLD.check_in_date, OLD.check_out_date, OLD.customer_id, r.room_number, r.price, r.amenities,
r.capacity, r.has_sea_view, r.has_mountain_view, r.extendable, OLD.booking_id
   FROM room r
   WHERE r.room_number = OLD.room_number;
END;
//
DELIMITER ;
```

Create a trigger to archive renting information when a renting is deleted:

```
DELIMITER //
CREATE TRIGGER archive_renting
AFTER DELETE ON renting
FOR EACH ROW
BEGIN
```

```
    INSERT INTO archive (check_in_date, check_out_date, customer_id, room_number, room_price,
room_amenities, room_capacity, room_has_sea_view, room_has_mountain_view, room_extendable, renting_id)
    SELECT OLD.check_in_date, OLD.check_out_date, OLD.customer_id, r.room_number, r.price, r.amenities,
r.capacity, r.has_sea_view, r.has_mountain_view, r.extendable, OLD.renting_id
    FROM room r
    WHERE r.room_number = OLD.room_number;
END;
//
DELIMITER ;
```

Trigger ensure check-out date is after check-in date in the booking table

```
DELIMITER //
CREATE TRIGGER ensure_checkout_after_checkin
BEFORE INSERT ON booking
FOR EACH ROW
BEGIN
    IF NEW.check_out_date <= NEW.check_in_date THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Check-out date must be after the check-in date.';
    END IF;
END;
//
CREATE TRIGGER ensure_checkout_after_checkin_update
BEFORE UPDATE ON booking
FOR EACH ROW
BEGIN
    IF NEW.check_out_date <= NEW.check_in_date THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Check-out date must be after the check-in date.';
    END IF;
END;
//
DELIMITER ;
```

Query 1: Find all hotels in a specific city

```
SELECT * FROM hotel WHERE city = 'Chicago';
```

Query 2: finding the total number of rooms in each hotel chain

```
SELECT h.hotelChain_id, SUM(h.number_rooms) as total_rooms
FROM hotel h
JOIN hotel_chain hc ON h.hotelChain_id = hc.hotelChain_id
GROUP BY h.hotelChain_id;
```

Query 3: Find all customers who registered before a specific date:

```
SELECT * FROM customer WHERE registration_date < '2023-04-13';
```

Query 4: Find all employees who work in a specific hotel chain:

```
SELECT * FROM employee e
JOIN hotel h ON e.hotel_id = h.hotel_id
JOIN hotel_chain hc ON h.hotelChain_id = hc.hotelChain_id
WHERE hc.hotelChain_id = 1;
```

Index and Explanation

1. Index on room(price)
CREATE INDEX idx_room_price ON room(price);

Explanation: This index is useful for efficiently searching and sorting rooms by price. It can help when trying to find rooms within a specific price range or sorting available rooms by price.

2.Index on hotel(categorization):
CREATE INDEX idx_hotel_categorization ON hotel(categorization);

Explanation: This index helps for searching and sorting hotels by their categorization (star rating). It can help users to find hotels that meet their quality preferences or to rank hotels based on their star rating.

3. Index on booking(check_in_date):
CREATE INDEX idx_booking_check_in_date ON booking(check_in_date);

4. Index on booking(check_out_date):
CREATE INDEX idx_booking_check_out_date ON booking(check_out_date);

Explanation for 3 and 4: These indexes are useful for searching and sorting bookings based on check-in and check-out dates. They can help in finding available bookings during specific periods or analyzing booking patterns over time.

5. Index on hotel(name):
CREATE INDEX idx_hotel_name ON hotel(name);
Explanation: This index can search and sort hotels by their names. It can help users find a specific hotel quickly or list hotels alphabetically.

View

View 1: Number of available rooms per area

```
CREATE OR REPLACE VIEW available_rooms_per_area AS
  SELECT
    h.city AS area,
    COUNT(r.room_number) AS available_rooms
  FROM
    hotel h
  JOIN
    room r ON h.hotel_id = r.hotel_id
  GROUP BY
    h.city;
```

View 2: Capacity of all the rooms of a specific hotel

```
CREATE VIEW hotel_room_capacity AS
SELECT
  h.hotel_id,
  h.name AS hotel_name,
  SUM(CASE
      WHEN r.capacity = 'Single' THEN 1
      WHEN r.capacity = 'Double' THEN 2
      ELSE 0
    END) AS total_capacity
FROM
```

```
    hotel h
JOIN
    room r ON h.hotel_id = r.hotel_id
GROUP BY
    h.hotel_id, h.name;
```